Lecture 3: Metric Spaces and Topology

Mathematical Analysis

1 Metric Spaces

Definition 1.1 (Metric). [def:metric] Let X be a set. A *metric*, or *distance function*, on X is a function $d : X \times X \to \mathbb{R}$ satisfying the following four axioms: nonnegativity, identity of indiscernibles, symmetry, and the triangle inequality

Axiom 1.1 (Nonnegativity). [axm:metric-nonnegativity] *If d is a metric on X, then*

$$\forall x, y \in X, \quad d(x, y) \geq 0$$

Axiom 1.2 (Identity of Indiscernibles). [axm:metric-separation] *If d is a metric on X, then*

$$\forall x, y \in X, \quad d(x, y) = 0 \iff x = y$$

Axiom 1.3 (Symmetry). [axm:metric-symmetry] *If d is a metric on X, then*

$$\forall x, y \in X, \quad d(x, y) = d(y, x)$$

Axiom 1.4 (Triangle Inequality). [axm:metric-triangle-inequality] *If d is a metric on X, then*

$$\forall x, y, z \in X, \quad d(x, z) \leq d(x, y) + d(y, z)$$

Definition 1.2 (Metric Space). [def:metric-space] A *metric space* is a pair (X, d), where X is a set and d is a metric on X

2 Open Balls and Neighbourhoods

Definition 2.1 (Open Balls). [def:open-balls] Let (X, d) be a metric space. The *open-ball generator* is

$$\text{OpenBall} : \mathbb{R}_+ \to \big(X \to \mathcal{P}(X)\big) \equiv r \mapsto x \mapsto \{y \in X : d(x, y) < r\}$$

For $r \in \mathbb{R}_+$ and $x \in X$, $\text{OpenBall}(r)(x)$ is the *open ball centred at x with radius r*

`LaTeX: \OpenBall (r)(x)`

Remark 2.1 (Open Ball Intuition). [rem:open-ball-intuition] An open ball in a metric space (X, d), centred at $x \in X$ with positive radius r, is the collection of all elements of X whose distance from x is less than r

Convention 2.1 (Open Ball Notation). [conv:open-ball-notation] For the open-ball generator, write

$$B_r(x) \equiv \text{OpenBall}(r)(x)$$

`LaTeX: \ball {r}{x}`

Definition 2.2 (Neighbourhood). [def:neighbourhood] Let (X, d) be a metric space, and let $x \in X$. A set $U \subseteq X$ is a *neighbourhood of x* if

$$\exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq U$$

Convention 2.2 (Radius Neighbourhood Notation). [conv:radius-neighbourhood-notation] In a metric space (X, d), some texts call the open ball $B_r(x)$, with $r \in \mathbb{R}_+$ and $x \in X$, the *r-neighbourhood of x* and write

$$V_r(x) \equiv B_r(x)$$

Lemma 2.1 (The Empty Set and the Whole Space). [lem:metric-empty-and-whole-open] *Let (X, d) be a metric space. Then*

$$\forall U \in \{\emptyset, X\},\ \forall x \in U,\ \exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq U$$

Proof. **Case $U = X$:** For every $x \in X$, choose $r = 1$. By definition, $B_1(x) \subseteq X$
Case $U = \emptyset$: There is no $x \in \emptyset$, so the universally quantified condition holds vacuously
The empty-set case does not require finding a ball contained in $\emptyset$, since there is no centre to consider □

Lemma 2.2 (Finite Intersections). [lem:metric-finite-intersections] *Let (X, d) be a metric space, and let $U, V \subseteq X$ satisfy*

$$\forall x \in U,\ \exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq U$$
$$\forall x \in V,\ \exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq V$$

Then

$$\forall x \in U \cap V,\ \exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq U \cap V$$

Proof. Let $x \in U \cap V$. By the assumptions, choose $r_U, r_V \in \mathbb{R}_+$ such that $B_{r_U}(x) \subseteq U$ and $B_{r_V}(x) \subseteq V$. Let $r = \min(r_U, r_V) > 0$. Then

$$B_r(x) \subseteq B_{r_U}(x) \subseteq U, \qquad B_r(x) \subseteq B_{r_V}(x) \subseteq V$$

Therefore, $B_r(x) \subseteq U \cap V$ □

Lemma 2.3 (Arbitrary Unions). [lem:metric-arbitrary-unions] *Let (X, d) be a metric space, and let $U : I \to \mathcal{P}(X)$ satisfy*

$$\forall i \in I,\ \forall x \in U(i),\ \exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq U(i)$$

Then

$$\forall x \in \bigcup_I U,\ \exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq \bigcup_I U$$

Proof. Let $x \in \bigcup_I U$. Choose $i \in I$ with $x \in U(i)$. By the assumption, choose $r \in \mathbb{R}_+$ such that

$$B_r(x) \subseteq U(i) \subseteq \bigcup_I U$$

□

3 Distance on an Ordered Field

Definition 3.1 (Distance). [def:distance] Let F be an ordered field. For $x, y \in F$, the *distance* between x and y is

$$d(x, y) \equiv |x - y|$$

`LaTeX: \dist {x}{y}`

Lemma 3.1 (Symmetry of Distance). [lem:distance-symmetry] *Let F be an ordered field. For all $x, y \in F$,*

$$d(x, y) = d(y, x)$$

Proof. By definition, $d(x, y) = |x - y|$ and $d(y, x) = |y - x| = |-(x - y)|$. It suffices to show $|-(x - y)| = |x - y|$; we split on the sign of $x - y$
Case 1: $x - y \geq 0$. Then $|x - y| = x - y$. Also $-(x - y) \leq 0$, so $|-(x - y)| = -(-(x - y)) = x - y = |x - y|$
Case 2: $x - y < 0$. Then $|x - y| = -(x - y)$. Also $-(x - y) > 0$, so $|-(x - y)| = -(x - y) = |x - y|$ □

Lemma 3.2 (Triangle Inequality for Distance)**.** [lem:triangle-inequality-distance] *Let F be an ordered field. For all $x, y, z \in F$,*

$$d(x, z) \leq d(x, y) + d(y, z)$$

Proof. Let $a = x - y$ and $b = y - z$. Then $x - z = a + b$, so $d(x, z) = |a + b|$. We split on the sign of $a + b$
Case 1: $a + b \geq 0$. Then $|a + b| = a + b$. Since $a \leq |a|$ and $b \leq |b|$,

$$|a + b| = a + b \leq |a| + |b| = d(x, y) + d(y, z)$$

Case 2: $a + b < 0$. Then $|a + b| = -(a + b) = -a - b$. Since $-a \leq |a|$ and $-b \leq |b|$,

$$|a + b| = -a - b \leq |a| + |b| = d(x, y) + d(y, z)$$

□

Corollary 3.1 (Metric on a Complete Ordered Field)**.** [cor:complete-ordered-field-metric] *Let F be a complete ordered field, and let d be the absolute-value distance on F. Under the identification with the reals, (F, d) is a metric space*

Proof. Under this identification, the distance is real-valued. By the definition of absolute value,

$$d(x, y) = |x - y| \geq 0, \qquad d(x, y) = 0 \iff x - y = 0 \iff x = y$$

Symmetry and the triangle inequality give the remaining two metric axioms □

Lemma 3.3 (ε-Criterion for Equality)**.** [lem:epsilon-equality] *Let F be an ordered field. For $a, b \in F$,*

$$a = b \iff \forall \varepsilon \in F_+,\ |a - b| < \varepsilon$$

Proof. In order to proof Epsilon Equality Lemma, we need to prove both directions
(a): $a = b \implies \forall \varepsilon \in F_+, |a - b| < \varepsilon$
Suppose $a = b$. Then $|a - b| = 0$. For any $\varepsilon > 0$, we have $|a - b| = 0 < \varepsilon$
(b): $\forall \varepsilon \in F_+, |a - b| < \varepsilon \implies a = b$
Assume $a \neq b$
Then, this means that

$$\exists \varepsilon_0 \in F_+, |a - b| = \varepsilon_0$$

Taking $\varepsilon = \varepsilon_0$ gives $\varepsilon_0 = |a - b| < \varepsilon_0$, a contradiction
So, we have $\forall \varepsilon \in F_+, |a - b| < \varepsilon \implies \neg a \neq b$
Since $\neg a \neq b \iff a = b$, So, $\forall \varepsilon \in F_+, |a - b| < \varepsilon \implies a = b$
Therefore, both implications hold, and hence proven □

4 Topological Spaces

Definition 4.1 (Topology)**.** [def:topology] Let X be a set. A *topology* on X, called $\mathcal{T}$, is a subset of $\mathcal{P}X$ satisfying containment of the empty set and the whole set, closure under finite intersections, and closure under arbitrary unions

Axiom 4.1. [axm:topology-empty-and-whole] *If $\mathcal{T}$ is a topology on X, then*

$$\emptyset, X \in \mathcal{T}$$

Axiom 4.2. [axm:topology-finite-intersections] *If $U, V \in \mathcal{T}$, then*

$$U \cap V \in \mathcal{T}$$

Axiom 4.3. [axm:topology-arbitrary-unions] *If $U : I \to \mathcal{T}$ is a family of sets in $\mathcal{T}$, then*

$$\bigcup_I U \in \mathcal{T}$$

Definition 4.2 (Topological Spaces)**.** [def:topological-space] A *topological space* is a pair $(X, \mathcal{T})$, where X is a set and $\mathcal{T}$ is a topology on X

Definition 4.3 (Induced Topology)**.** [def:induced-topology] Let (X, d) be a metric space, where we define a *induced topology* $\mathcal{T}_d$ induced by d ;s

$$\mathcal{T}_d \equiv \{U \in \mathcal{P}(X) \mid \forall x \in U,\, \exists r \in \mathbb{R}_+,\, B_r(x) \subseteq U\}$$

Theorem 4.1 (Induced Topology is a Topology)**.** [thm:induced-topology-is-a-topology]

$$(X, d) \text{ is a metric space} \implies (X, \mathcal{T}_d) \text{ is a topological space}$$

Proof. **TODO:** Apply the three metric-space lemmas to verify the topology axioms □

5 Open and Closed Sets

Definition 5.1 (Open Sets)**.** [def:open-sets] A set $U \subseteq X$ is *open* in a topological space $(X, \mathcal{T})$ if $U \in \mathcal{T}$.

Definition 5.2 (Closed Sets)**.** [def:closed-sets] A set $C \subseteq X$ is *closed* in a topological space $(X, \mathcal{T})$ if its complement $X \setminus C$ is open, i.e., $X \setminus C \in \mathcal{T}$.

Definition 5.3 (Limit Points)**.** [def:limit-points] A *limit point* $x : X$ of a set $U \subseteq X$ is a point that

$$\forall \varepsilon \in \mathbb{R}_+,\, B_\varepsilon(x) \cap U \neq \emptyset$$

6 Compact Sets

7 Heine-Borel Theorem

8 Cantor Sets